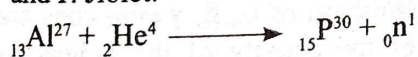


Chapter Highlights

Terms, Reasons of radioactivity, Units, Features of α , β and γ rays, Group displacement law, Radioactive disintegration series, Kinetics of radioactive disintegration, Half life, Average life, Nuclear fission and fusion, Nuclear reactions, Uses of radioactive isotopes and various levels of multiple-choice questions.

RADIOACTIVITY AND NUCLEAR REACTIONS

- The phenomenon of natural radioactivity and natural disintegration was discovered by Henri Becquerel.
- Radioactivity was first observed in potassium uranyl sulphate $K_2UO_2(SO_4)_2$.
- α and β rays were discovered by Rutherford while γ rays were discovered by Villiard.
- Artificial radioactivity was introduced by I. Curie and F. Jiolet.



- M. Curie and P. Curie extracted Po, Ra from pitch blende (uranium ore) and introduced the word radioactive.
- Ra shows maximum radioactivity.
- Radioactivity in thorium was discovered by G. C. Schmidt.
- Isotope with higher mass number shows radioactivity, for example, ${}_1H^3$, ${}_6C^{14}$, ${}_8O^{16}$
- Last elements of most of the groups shows radioactivity.
- Fr is the only radioactive element found in liquid state.
- γ rays has 10^{10} times more penetration power than α rays.
- A radioactivity change is irreversible and liberates 1 million times more energy than a chemical change.

- External factors (temperature, pressure etc.) do not affect radioactivity.
- Radioactivity does not need activation energy.
- α , β , γ ray emission can be easily detected as they cause luminescence on the ZnS screen placed in their path.

RADIOACTIVITY

- Henri Becquerel observed that uranium minerals emit some peculiar rays which can effect a photographic plate even in dark. This property was called radioactivity and these rays were called radioactive rays.
- It is a nuclear property shown by atoms with unstable nucleus like U, Th, Ra.

Reasons for Unstable Nucleus or Radioactivity

1. n/p ratio > 1.5 (Here, β emission occurs to decrease n/p ratio)
2. Radioactive substances have a n/p ratio < 1 (Here, α emission or positron emission or K-shell electron capture occurs to increase n/p ratio).

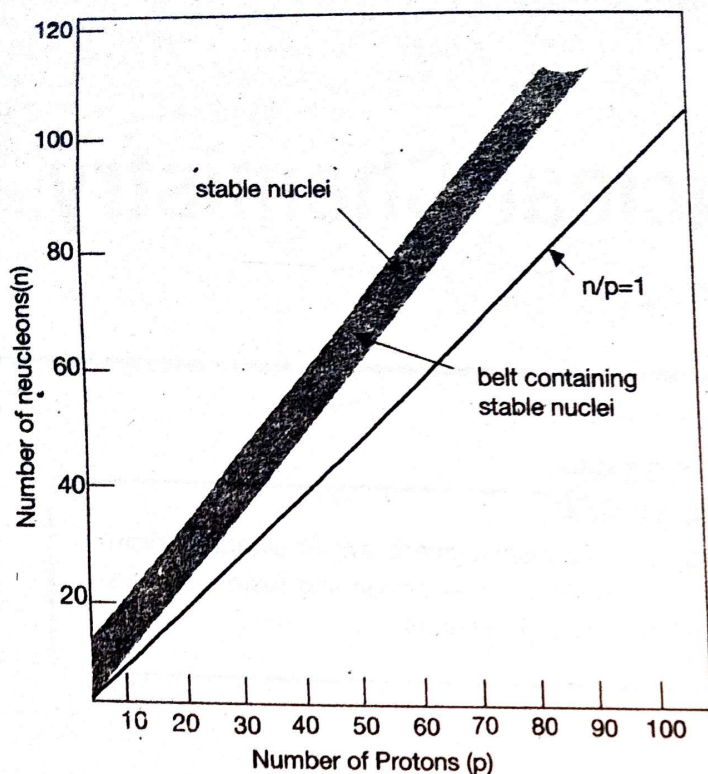


Figure 21.1 Zone of Stability on the Basis of n/p Ratio

- Radioactive substances have a low value of binding energy. The stability of a nucleus is measured directly in terms of the binding energy per nucleon ($\bar{B}$), which may be defined as the amount of energy released when the total number of nucleons (protons and neutrons) combine together to give the nucleus, divided by the number of nucleons.
- Radioactive substances have a high value of packing fraction—positive value of packing fraction implies an unstable nucleus.

K Electron Capture

Here, during a nuclear reaction, the nucleus of one of the atoms captures an electron from K-shell which converts proton to neutron. It is common in nuclei having $Z > 70$. For example,



Mass Defect (Δm) and Binding Energy (B) and Packing Fraction (f)

- $\Delta m = \text{Isotopic mass} - \text{Mass number}$
- Packing Fraction (f) = $\frac{\Delta m}{A} \times 10^4$
Here, A is mass number or atomic weight.
Stability $\propto \frac{1}{f}$ value
Hence, positive value means instability.

3. Binding energy is given as

$$B = \Delta m \times 931.5 \text{ MeV}$$

Hence, binding energy is the energy equivalent of the mass defect.

$$\bar{B} = \frac{BE}{\text{No. of Nucleon}}$$

Here, $\bar{B}$ is binding energy per nucleon.

Magic Numbers

Those elements which have 2, 8, 20, 28, 50, 82, 114 number of protons or 2, 8, 20, 28, 50, 82, 114, 126, 184 number of neutrons are quite stable and abundant in nature. These numbers are called magic numbers and the elements having these numbers do not show radioactivity.

Units of Radioactivity

$$1 \text{ Curie} = 3.7 \times 10^{10} \text{ dis/s}$$

$$1 \text{ Rutherford} = 10^6 \text{ dis/s}$$

$$1 \text{ Cr} = 3.7 \times 10^4 \text{ Rutherford}$$

$$1 \text{ Bq} = 1 \text{ dis/s}$$

$$1 \text{ Gray (gy)} = 1 \text{ kg tissue receiving one Joule of energy}$$

Measurement and Detection of Radioactivity

As radioactive emissions ionize the gases, so the degree of ionization of a gas is taken as a measurement of the rate of emission of α , β , γ emissions and also as a measure of the activity of the radioactive substance.

It is measured by one these methods:

- Geiger Muller counter method
- Wilson Cloud chamber method
- Dosimeters

Nature of Radiations

Rutherford studied the properties of these radiations and these can be summarized as shown in Table 21.1 ahead.



- α particle emission occurs only when Z (atomic number) is greater than 82.
- β particle emission occurs when neutron changes into proton
 ${}_0^1n^1 \longrightarrow {}_1^1p^1 + {}_{-1}^0\beta^0$
- When a proton changes into neutron, positron emission occurs.
 ${}_1^1p^1 \longrightarrow {}_0^1n^1 + {}_{+1}^0\beta^0$

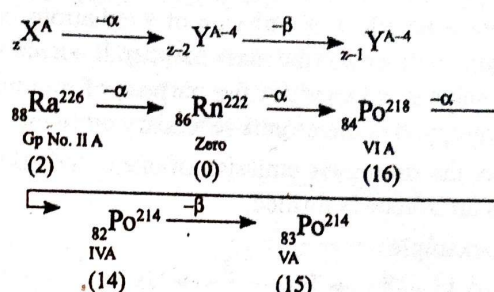
Group Displacement Law

Group displacement law was introduced by Soddy, Fajans and Russell. According to this law:

1. Only one type of particle is emitted at a time
2. When an α particle is emitted, the newly formed element will have an atomic number less by two units. The mass number reduces by four units while the group position is shifted two places towards the left.

3. When a β particle is emitted the newly formed element will have an atomic number increased by one unit, while the mass number remains the same and the group position moves one position towards the right.

For example,

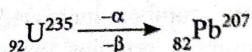


- Number of α particles = $\frac{\text{Loss in mass number}}{4}$
 Number of β particles = $2 \times \alpha - [Z_1 - Z_2]$
 Here, Z_1 = Atomic number of parent element
 Z_2 = Atomic number of daughter element.

Table 21.1 Comparison of Properties of Various Radioactive Radiations

Properties	α particles or α rays	β particles or β rays	γ rays
Representation	It is shown as a helium nucleus or helium ions ${}_2\text{He}^4$ or He^{++} .	It is shown as an electron ${}_{-1}^0e^0$	It is shown as ${}_0^0\gamma^0$.
Charge and Mass	It carries 2 unit positive charge and 4 unit mass.	It carries 1 unit negative charge and no mass.	These are electro magnetic rays with very short wavelength (0.05 Å)
Action of magnetic field	These are deflected towards the cathode.	These are deflected towards the anode.	These are not deflected.
Velocity	2×10^9 cm/s or 2×10^7 m/s (1/10th of that of light)	2.36 to 2.83×10^{10} cm/s (2.36 to 2.83×10^8 m/s)	Same as that of light 3×10^{10} cm/s (3×10^8 m/s)
Ionizing power	Very high; nearly 100 times more than that of β rays.	Low; nearly 100 times more than that of γ rays.	Very low
Penetrating power	Low	100 times that of α particles	1000 times that of β particles
Effect of ZnS plate	They cause luminescence	Very little effect	Very little effect
Range	Very small (8–12 cm)	More than of α particles	More than of β particles
Nature of product	Product obtained by the loss of one α particle has atomic number less by 2 units and mass number less by 4 units.	Product obtained by the loss of one β particle has atomic number more by one unit, without any change in mass number.	There is no change in the atomic number as well as in mass number.

For example,

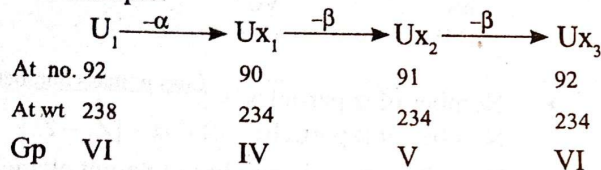


Number of α particles = $\frac{235-207}{4} = 7$

Number of β particles = $2 \times 7 - (92 - 82) = 4$

- There is no effect of emission of γ radiations on the atomic number and the mass number. It occurs when the nucleus is excited for the purpose of minimizing its energy. It is therefore a secondary emission.
- After the successive emission of one α and 2β particles an isotope is formed.

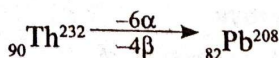
For example:



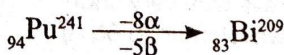
- In case of lanthanides and actinides, the group position does not change after emission of α or β particles that is, it remains IIIB.

Radioactive Disintegration Series

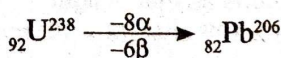
4n (Thorium) series Here, mass number divides completely by four.



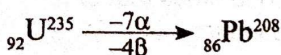
4n + 1 (Neptunium) series Here, the remainder is one when the mass number is divided by four. It is an artificial series and here the end product is not lead.



4n + 2 (Uranium) series Here, the remainder is two when the mass number is divided by four.



4n + 3 (Actinium) series Here, the remainder is three when the mass number is divided by four.



Rate of Radioactive Disintegration and Disintegration Constant

The decay of a radioactive isotope takes place by the disintegration of atomic nucleus. Rate of reaction does not depend upon external factors like temperature, pressure and amount or mass of substance.

$$\text{Rate of reaction} = -\frac{dN}{dt}$$

$$-\frac{dN}{dt} \propto N$$

[N = Number of undecayed atoms of the isotope present in the given amount of sample.]

$$-\frac{dN}{dt} = \lambda N$$

Here, λ or k is decay constant.

$$N = N_0 e^{-\lambda t}$$

So, λ or $k = \frac{2.303}{t} \log_{10} \frac{a}{a-x}$ or $\frac{N_0}{N}$

Here, N_0 or a is the initial amount or activity while $a-x$ or N is the activity or amount left after the time t .

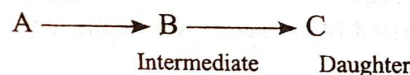
- λ depends only on half-life period ($t_{1/2}$). It does not depend upon the mass or amount of substance.

$$\lambda \propto \frac{1}{t_{1/2}}$$

$$\lambda \text{ or } k = \frac{0.693}{t_{1/2}}$$

Radioactive Equilibrium or Steady State

Radioactive equilibrium is obtained when the rate of formation of an element and its further disintegrations are the same.



$$\lambda_1 N_1 = \lambda_2 N_2$$

$$\frac{N_1}{N_2} = \frac{\lambda_2}{\lambda_1} = \frac{(t_{1/2})_1}{(t_{1/2})_2}$$

$$\frac{N_2}{N_1} = \frac{\lambda_1}{\lambda_2 - \lambda_1}$$

It is a transient equation which is infact a steady state.

Half-life Period ($t_{1/2}$)

Half-life period is the duration in which a radioactive substance disintegrates half of its initial amount.

- It does not depend upon the mass, physical or chemical change and external factors.

$$t_{1/2} = \frac{0.693}{k \text{ or } \lambda}$$

$$T = n \times t_{1/2}$$

$$N = N_0 \times \left(\frac{1}{2}\right)^n$$

Here, n = Number of disintegrations

Average Life

Average life is the reciprocal of disintegration constant λ .

$$\tau = \frac{1}{k}$$

$$\tau = 1.44 \times t_{1/2}$$

FACTS TO REMEMBER

Rate of disintegration or activity is given as

$$-\frac{dN}{dt} = \frac{0.693}{T} \times \frac{w(g)}{\text{mass no.}} \times 6.023 \times 10^{23}$$

$$\text{Amount decayed after } n = \frac{N_0 [2^n - 1]}{2^n}$$

NUCLEAR REACTIONS

- The reactions taking place between atoms, involving nuclear changes, are called nuclear reactions.
- The first artificial nuclear reaction or transition was carried out by Rutherford.

$${}_7\text{N}^{14} + {}_2\text{He}^4 \longrightarrow {}_8\text{O}^{17} + {}_1\text{H}^1$$
- The nucleus that gets bombarded by a radioactive particle is known as **target nucleus**.
- The particle that is used to bombard the target nucleus is known as a **projectile**.
- The nuclide which is formed in the nuclear reaction is known as a **product nuclide**.
- The other particle produced in the nuclear reaction is known as **ejected particle**.
- The projectiles and ejectiles in a nuclear reactions are always shown by their symbols instead of nuclides. For example, ${}_7\text{N}^{14} (\alpha, \beta) {}_8\text{O}^{17}$
- Energy change of nuclear reaction is shown on the products side by $-Q$ if it is endoergic (endothermic) or $+Q$ if it is exoergic (exothermic).

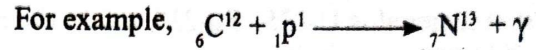
Difference Between a Nuclear and a Chemical Reaction

- In a nuclear reaction the number of the nucleons of the target nuclide change while in chemical reactions nuclei are unaffected as only electrons take part in it.
- Transformation and transmutation of an atom occurs during a nuclear reaction but not in a chemical reaction.
- There is a loss of mass during nuclear reactions but not in chemical reactions.
- A nuclear reaction does not depend upon external factors while the chemical reaction depends upon external factors.
- In a nuclear reaction much more energy is released than in a chemical reaction.
- During a nuclear reaction, emission of α or β particles takes place but this does not occur a chemical reaction.

Types of Nuclear Reactions

Nuclear reactions can be classified in the following ways:

- Depending on the overall change in atoms and energy.
- Depending on the nature of projectile particles.
- When the projectile particle is absorbed by the target nucleus with the emission of γ ray, it is known as **projectile capture reaction**.

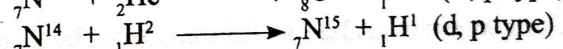
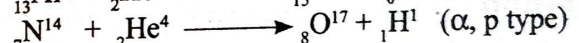
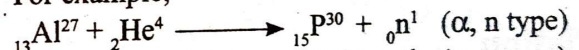


- When the projectile particle is absorbed by the target nucleus and a new nucleus is formed with the emission of a new particle it is called particle-particle reaction. It is of following types:

1. Charged p article reaction:

Projectile	Type
p	(p, n) (p, α) (p, d)
${}_1\text{H}^2$	(d, p) (d, n) (d, α)
α	(α , n) (α , p)

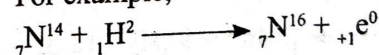
For example,



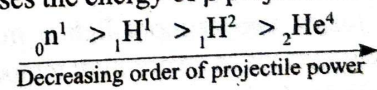
2. Neutral particle reaction

Projectile	Type
n	(n, p) (n, α)

For example,



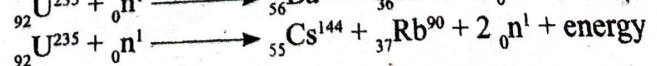
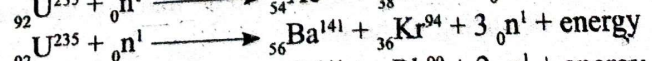
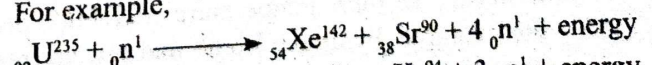
- (a) The energy of positive particles is increased by a machine like cyclotron. Betatron or synchrotron increases the energy of β projectiles. For example,



Nuclear Fission

- Nuclear fission was introduced by Otto Hahn and Fritz Strassman.
- It is the splitting of heavy nuclei into two or more lighter nuclei with a release of huge amount of energy in the form of nuclear energy. This energy is a result of the mass defect.

For example,



- It occurs at a lower temperature than that for fusion.
- Lesser energy is released than that released in fusion.
- It is measured by liquid drop method.
- It is a chain reaction which can further be of two types

Controlled chain reaction

Here, chain reaction is controlled. For example, nuclear reactions carried out in a nuclear reactor.

Facts about Nuclear Reactor

- Fission material is U-235, Pu-239 and fertile material is U-238.
- Here, graphite, cadmium, berilium, boron rods act as neutron absorber.
- Here, heavy water works as moderator to slow down the speed of neutrons.
- Heavy water, polyphenyls or graphite acts as a coolant to remove excess heat from the nuclear reactor.

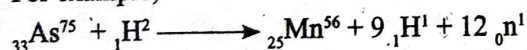
Uncontrolled chain reaction

Here, chain reaction cannot be controlled as is the case of nuclear reactions in an atom bomb fueled by U-235 or Pu-239.

Spallation

- The concept of spallation was introduced by Seaborg.
- It is the splitting of heavy nuclei into lighter nuclei by high energy particles. Here, 10 to 20 neutrons are released.

For example,

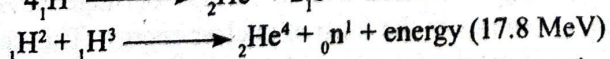
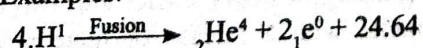


Nuclear Fusion

In nuclear fusion two or more lighter nuclei combine to form a heavy nucleus at a very high temperature ($>10^6$ K) with the release of a huge amount of energy in the form of nuclear energy. Here, released energy is due to the mass defect.

- Nuclear fusion is an exoergic reaction.

Examples:



- It can not be controlled and it is not a chain reaction.
- It occurs at high temperature and high pressure. For example, hydrogen bomb and nuclear reactions occurring in the sun.

Uses of Artificial Radioactive Isotope

Carbon dating

It was introduced by W. S. Libby to find the age of fossils and old wood with the help of C-14.

The principle reaction involved is given by the reaction.



The ratio of C-14/C-12 in living things is $1/10^{12}$. There is one atom of C-14 for every 7.49×10^{11} carbon atom in CO_2 of the atmosphere and in all living species.

$${}_6\text{C}^{14} \longrightarrow {}_7\text{N}^{14} + {}_{-1}\text{e}^0$$

$$t = \frac{2303}{k \text{ or } \lambda} \log_{10} \frac{N_0}{N}$$

$$t = \frac{2303 \times t_{1/2}}{0.693} \log_{10} \frac{N_0}{N}$$

Here, $t_{1/2}$ = Half-life of C-14

N_0 = Original activity or amount or ratio of C-14/C-12 in fresh sample

N = Final activity or amount or ratio of C-14/C-12 in old or dead sample

Rock dating

It is based on the kinetics of radioactive decay. Here, it is assumed that all the lead present in the sample is due to decay of U-238. It is used to find the age of rocks, earth, minerals using U-238, Pb-206

Here, N_0 = U-238 + Pb-206, N = U-238

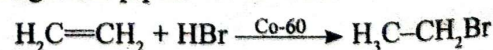
$$\text{Hence, } t = \frac{2303}{K \text{ or } \lambda} \log \frac{N_0}{N}$$

In medicines

- Na-24 (γ) is used to detect tumours and blood clots.
- I-131 (β) is used in the treatment of thyroid gland.
- Co-50 (γ) is used to check cancer growth.
- Co-60(γ) + Au-198(γ) is used to check cancer and tumour treatment.
- P-30 (β) is used in the treatment of skin diseases.
- P-32 is used in the treatment of leukemia.

In industry

Sb-124 is used to study the flow of petroleum through underground pipes.



In agriculture

P-32 is used to trace the assimilation of phosphorus by plants.

Half-life, Rate Constant

7. Half-life period of iodine (atomic mass 125) is 60 days. Calculate the % of radioactivity remaining after 120 days.

Solution

$$t_{1/2} = 60 \text{ days as } t = 120 \text{ days}$$

$$\text{As } n = \frac{t}{t_{1/2}} = \frac{120}{60} = 2$$

$$\text{So, \% of radioactivity remaining after 2 half-lives} \\ = \frac{N_0}{2^2} = \frac{100}{4} = 25\%$$

8. A sample of radioactive I^{133} gave with a GM counter 3150 counts per minute at a certain time and 3055 counts per minute exactly one hour later. What is the half-life period of I^{133} ?

Solution

$$k = \frac{2.303}{t} \log_{10} \frac{N_0}{N} = \frac{2.303}{1} \log_{10} \frac{3150}{3055} = 3.06 \times 10^{-2} \text{ year}^{-1}$$

$$t_{1/2} = \frac{0.693}{3.06 \times 10^{-2}} = 22.64 \text{ year}$$

9. Calculate the time required to lose radioactive intensity of a radioactive sample of Na^{24} to 70% of its original value. Given $t_{1/2}$ of Na^{24} = 14.8 hours.

Solution

$$\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{14.8} = 4.68 \times 10^{-2} \text{ hour}^{-1}$$

$$t = \frac{2.303}{\lambda} \log_{10} \frac{N_0}{N}$$

$$= \frac{2.303}{4.68 \times 10^{-2}} \log_{10} \frac{100}{30} = 25.73 \text{ hour}$$

10. At a certain instant, a piece of radioactive material contains 10^{12} atoms. The half-life of material is 30 days. Calculate the number of disintegrations in the first second.

Solution

$$k = \frac{0.693}{t_{1/2}}$$

$$t_{1/2} = 30 \text{ day}$$

$$k = \frac{0.693}{30 \times 24 \times 60 \times 60} \text{ sec}^{-1}$$

$$\text{As } N = 10^{11} \text{ atoms}$$

$$-\frac{dN}{dt} = kN$$

$$-\frac{dN}{dt} = \frac{0.693 \times 10^{11}}{30 \times 24 \times 60 \times 60} \text{ sec}^{-1}$$

$$= 2.67 \times 10^5 \text{ sec}^{-1}$$

11. The activity of a radioactive isotope falls to 12.5% in 90 days. Calculate the half-life and decay constant of the radio isotope.

Solution

$$k = \frac{2.303}{t} \log_{10} \frac{N_0}{N}$$

$$k = \frac{2.303}{90} \log_{10} \frac{100}{12.5}$$

$$k = 23.03 \times 10^{-3} \text{ day}^{-1}$$

$$t_{1/2} = \frac{0.693}{23.03 \times 10^{-3}} = 30 \text{ days}$$

12. One of the hazards of nuclear explosion is the generation of Sr^{90} and its subsequent incorporation in bones. This nucleide has a half-life of 28.1 years. Suppose, one microgram was absorbed by a new born child, how much Sr^{90} will remain in his bones after 20 years?

Solution

$$t_{1/2} = 28.1 \text{ year}$$

$$N_0 = 10^{-6} \text{ g}$$

$$t = 20 \text{ year}$$

$$N = ?$$

$$t = \frac{2.303}{k} \log_{10} \frac{N_0}{N}$$

$$\therefore t = \frac{0.693}{t_{1/2}} = \frac{0.693}{28.1}$$

$$t = \frac{2.303 \times 28.1}{0.693} \times \log_{10} \frac{10^{-6}}{N}$$

$$N = 6.1 \times 10^{-2} \text{ g}$$

Carbon Dating, Rock Dating

13. A sample of U^{238} (half-life = 4.5×10^9 year) ore is found to contain 23.8 g of U^{238} and 20.6 g of Pb^{206} . Calculate the age of the ore.

Solution

$$N = \text{U present} = \frac{23.8}{238} = 0.1 \text{ g atom}$$

$$N_0 = \text{U present} + \text{U decayed}$$

$$= \frac{23.8}{238} + \frac{20.6}{206} = 0.1 + 0.1 = 0.2 \text{ g atoms}$$

$$\lambda = \frac{0.693}{4.5 \times 10^9} = 1.54 \times 10^{-10} \text{ Year}^{-1}$$

$$t = \frac{2.303}{1.54 \times 10^{-10}} \log \frac{0.2}{0.1} = 4.5 \times 10^9 \text{ Year}$$

14. The bones of a prehistoric bison were found to have a C^{14} activity of 2.80 dis/min g carbon. How long ago, approximately did the animal live? Given $A_0 = 15.3$ dis/min g, $t_{1/2} = 5730$ years.

Solution

$$\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{5730} = 1.209 \times 10^{-4} \text{ year}^{-1}$$

$$t = \frac{2.303}{1.209 \times 10^{-4}} \log_{10} \frac{153}{2.80} = 14049 \text{ Year}$$

15. A freshly cut piece of wood gives 16100 counts of β ray emission per minute per kg and an old wooden bowl gives 13200 counts per minute per kg. Calculate the age of wooden bowl. The half-life period of C^{14} is 5568 years.

Solution

$$\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{5568} = 1.244 \times 10^{-4} \text{ year}^{-1}$$

$$t = \frac{2.303}{\lambda} \log_{10} \frac{N_0}{N}$$

$$= \frac{2.303}{1.244 \times 10^{-4}} \log_{10} \frac{16100}{13200} = 1596 \text{ Years}$$

16. An old wooden chair shows a C^{14} activity which is 80% of the activity found today. Calculate the age of the sample. ($t_{1/2}$ of C^{14} = 5770 years).

Solution

$$t_{1/2} \text{ of } C^{14} = 5770 \text{ years}$$

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{5770} \text{ year}^{-1}$$

$$k = \frac{2.303}{t} \log_{10} \frac{N_0}{N}$$

$$\frac{0.693}{5770} = \frac{2.303}{t} \log_{10} \frac{100}{80}$$

$$t = 1858.25 \text{ years}$$

17. The half-life period of C^{14} is 5760 year. A piece of wood when buried in the earth had 1% C^{14} , now as charcoal, it has only 0.25% C^{14} . How long has the piece of wood been buried?

Solution

$$t = \frac{0.693}{t_{1/2}} = \frac{0.693}{5760} \text{ year}^{-1}$$

$$k = \frac{2.303}{t} \log_{10} \frac{1}{0.25}$$

$$\frac{0.693}{5760} = \frac{2.303}{t} \log_{10} 4 = \frac{2.303}{t} \times 2 \times 0.3010$$

$$t = \frac{2.303 \times 0.6020 \times 5760}{0.693} = 1152337 \text{ Years}$$



TOOL BOX AND SUMMARY OF IMPORTANT RELATIONS

Average Atomic Mass of Isotopes

$$= \frac{\% \times \text{At. wt of A} + \% \times \text{At. wt of B}}{100}$$

- To find number of α , β particles

$$\alpha \text{ particles} = \frac{\text{Change in mass number}}{4}$$

$$\beta \text{ particles} = 2 \times \alpha \text{ particles} - (Z_1 - Z_2)$$

Here, Z_1 = Atomic number of parent element

Z_2 = Atomic number of daughter elements

Radioactive Disintegration Constant

$$\lambda \text{ or } k = \frac{2.303}{t} \log \frac{a}{a-x} \text{ or } \frac{N_0}{N}$$

Here N_0 or a is initial amount or activity while $a - x$ or N is activity or amount left after time t .

Radioactive Equilibrium

$$\lambda_1 N_1 = \lambda_2 N_2$$

$$\frac{N_a}{N_b} = \frac{\lambda_a}{\lambda_a - \lambda_b}$$

Mass Defect (Δm) and Binding (B) Energy

Δm = Isotopic Mass - Mass number

$$\text{Packing Fraction (f)} = \frac{\Delta m}{A} \times 10^4$$

$$\text{Binding energy (B)} = \Delta m \times 931.6 \text{ MeV}$$

$$\bar{B} = \frac{B}{\text{Neutron (n + p)}}$$

Here, $\bar{B}$ is binding energy per nucleon.

Half-life Period ($t_{1/2}$)

$$t_{1/2} = \frac{0.693}{k \text{ or } \lambda}$$

Here, n = Number of disintegrations.

Average Life

$$\tau = \frac{1}{k}$$

Rate of Disintegration

$$\frac{-dN}{dt} = \frac{0.693}{t} \times \frac{W(g)}{\text{mass no.}} \times 6.023 \times 10^{23}$$

$$\text{Amount decayed after } n = \frac{N_0 2^n - 1}{2^n}$$

Carbon Dating Method

$$t = \frac{2.303}{k \text{ or } \lambda} \log_{10} \frac{N_0}{N}$$

$$t = \frac{2.303 \times t}{k \text{ or } \lambda} \log_{10} \frac{N_0}{N}$$

Here, $t_{1/2}$ = Half-life of C-14.

N_0 = Original activity or amount or ratio of C-14/C-12 in fresh sample

N = Final activity or amount or ratio of C-14/C-12, in an old or dead sample

Rock Dating Method

$$t = \frac{2.303 \times t}{k \text{ or } \lambda} \log_{10} \frac{N_0}{N}$$

$$t = \frac{2.303 \times t_{1/2}}{0.693 \text{ or } \lambda} \log_{10} \frac{N_0}{N}$$

Here, $t_{1/2}$ = Half-life of U-238

N_0 = Moles of U-238 + moles of Pb-206

N = Moles of U-238

CONCEPTS AT A GLANCE

Cobalt Bomb

It is a hydrogen bomb enclosed in a sheath of cobalt.

- ${}^3_1\text{H}$ is used as tracer in absorption of water by plants roots.
- Na^{24} is used to find leakage in pipes.
- ${}^{94}_{24}\text{Pu}$ is used in atom bomb with U-238 as fuel.

Neutrino

It explains conservation of any momentum in β -decay and is postulated to balance the species.

Anti Neutrino

It balances the spin to distinguish e^+ or e^- decays.

Harkin's Rule

According to this element of even atomic number (Z) are more stable and abundant than neighbouring element of odd 'Z'.

- Nuclear radius (r) of any atom is related to its mass number A as $r = R_0 A^{1/3}$

where $R_0 = \text{constant} = 1.4 \times 10^{-15} \text{ m}$.

- Area of cross section of nucleus is expressed in barns (1 barn = 10^{-24} cm^2).

Capture Reaction

If the particle used for bombardment is captured by the nucleus without release of any subsidiary particle, it is called capture reaction

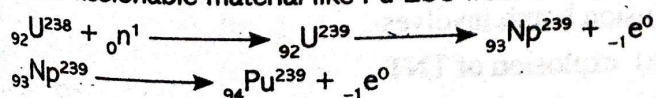
For example, ${}_3\text{Li}^7 + {}_2\text{He}^4 \longrightarrow {}_5\text{B}^{11} + \gamma$

Critical Mass

The minimum mass which the fissionable material must have for fission so that chain reaction continues at a constant rate is called critical mass. For U-235, it is 1–100 kg.

Breeder Reactors

Breeder reactors are those reactors in which the neutrons produced from fission of U-235 are partly used to carry on the fission of U-235 and partly used to produce some other fissionable material like Pu-239 from U-238.

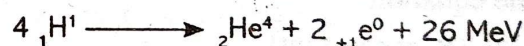
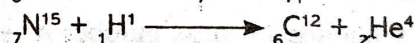
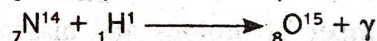
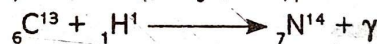
**Neutron Activation Analysis**

It is used for finding the trace of one element present in another by activating the trace element by bombarding with neutrons, for example, traces of Ag present in a lead painting.

- Activity of a nucleus is inversely proportional to its half-life or average life. Thus, shorter the half-life of an element, greater is its radioactivity, that is, higher number of atoms disintegrating per second. It is called as **Geiger-Nuttall rule**.
- The total life span of any radioactive element is infinite.
- Nuclear forces are 10^{21} times stronger than electrostatic forces.
- Every second, the sun loses $4.3 \times 10^9 \text{ kg}$ of mass by fusion reactions and that is a very huge amount of energy.

Carbon-Nitrogen Cycle

According to Bethe, this cycle is responsible for the origin of energy in stars.



But according to modern view, proton-proton interaction leading to the formation helium takes place as follows:

